Driver Drowsiness Alarm System

Digital Logic Design Project Report

Introduction

Driver drowsiness is considered one of the leading causes of road accidents worldwide. This project presents a Driver Drowsiness Alarm System designed using Arduino and digital logic concepts. The system monitors driver movement using a PIR sensor and activates warning alerts when inactivity is detected.

Project Objectives

1. Detect driver inactivity and drowsiness.
2. Trigger an alarm when inactivity exceeds a limit.
3. Reset the system when movement is detected.
4. Improve road safety.
5. Apply digital logic and embedded system concepts.

Project Value

The system provides an early warning mechanism that helps reduce accidents caused by fatigue and distraction. It demonstrates the use of embedded systems in smart safety applications.

System Idea

The system continuously monitors the driver's movement using a PIR sensor. If no movement is detected for a certain period, the alarm activates using an LED and buzzer. The system resets when movement is detected again.

Working Mechanism

- PIR sensor monitors movement.
- Timer starts when no movement is detected.
- Alarm activates after inactivity exceeds the limit.
- LED and buzzer warn the driver.
- Reset button restores normal operation.

Digital Logic Concepts Used

The project uses clock signals, counters, comparators, and flip-flops to measure inactivity time and manage the alarm state.

Future Improvements

• Add LCD status display.
• Add camera-based eye detection.
• Connect the system to vehicle controls.
• Add mobile notifications.

Conclusion

The Driver Drowsiness Alarm System is a practical embedded safety solution that combines Arduino, sensors, and digital logic concepts to improve driver awareness and reduce accidents.

Components Used

Component	Function
Arduino Uno	Main controller of the system
PIR Sensor	Detects driver movement
LED	Visual warning indicator
Buzzer	Sound alarm for warning
Push Button	Resets the system
Breadboard	Circuit connections
Jumper Wires	Connect electronic components